Report Task #01

Team Sigma

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QUESTION #01 :

How can we improve the accuracy and real-world reliability of rockburst intensity prediction models by integrating multimodal sensor data from multiple underground sites?"
This question captures:
The problem (limited, unrepresentative datasets),
The goal (improved accuracy, generalization),
The method (field sensor integration + augmentation + classical & deep learning).

ANSWER :

To build a robust rockburst prediction system without real-world datasets, we created a complete classical machine learning pipeline using synthetic and structured lab-based data, and deployed it with a Streamlit dashboard for interaction and visualization. Here's what was done:

1. **Problem Definition**

We addressed the issue of poor model generalization due to limited and single-source datasets. Our goal was to simulate a predictive model pipeline for rockburst intensity using multimodal features from seismic, acoustic, and geomechanical sources.

2. **Data Preparation**

We worked with structured features like:

Seismic: peak ground acceleration, b-value, event rate.

Acoustic: signal energy, FFT-based dominant frequencies.

Geomechanical: strength, brittleness, elastic strain energy.

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peak_accel	event_rate	b_value	signal_energy	duration	dominant_freq	brittleness	stress	elastic_energy	compressive_strength	tensile_strength	grade
0.812575	2	1.628524	131	5	165	0.450684	79	241	53	8	1
0.752866	3	1.559038	249	13	168	0.497278	62	269	48	8	2
0.810998	2	1.661275	203	5	158	0.470248	45	138	58	9	1
0.639434	1	1.680068	184	9	170	0.423906	60	174	32	11	0
0.771169	4	1.688472	278	10	182	0.474761	64	252	46	8	3
0.61194	2	1.902932	251	11	158	0.488023	48	130	37	8	1
1.858028	4	1.644985	329	8	162	0.491679	67	111	52	10	4
0.734889	2	1.771776	203	12	188	0.41169	35	166	38	10	1
0.728252	3	1.720536	110	12	161	0.452333	83	140	33	8	1
1.986537	3	1.868112	248	5	172	0.47241	45	297	37	8	2
0.503313	1	1.599421	235	7	184	0.477127	34	140	57	10	1
1.585239	3	1.696229	416	7	184	0.495093	86	189	43	8	4
0.642021	1	1.693001	233	8	155	0.479818	48	288	41	8	1
1.059353	2	1.884459	181	12	154	0.498003	50	293	42	9	1
1.901132	2	1.5122	269	6	179	0.400024	42	240	47	11	2
0.555079	1	1.520384	289	8	154	0.447417	45	184	38	11	1
0.973068	1	1.949424	214	5	155	0.467073	62	286	54	9	1
1.842834	3	1.847298	466	13	166	0.403086	68	172	42	9	4
0.869008	2	1.554827	221	11	188	0.432232	63	239	54	11	1
1.257689	2	1.689961	895	5	162	0.478486	66	178	52	9	4
1.052523	1	1.501298	135	5	157	0.474626	49	146	53	9	1
1.044087	1	1.613966	289	13	162	0.488073	36	297	37	10	1
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

3. **Preprocessing included:**

StandardScaler for feature normalization.

PCA to reduce dimensionality and handle multicollinearity (checked via VIF).

```
                 feature        VIF
0             peak_accel  10.234636
1             event_rate   5.923445
2                b_value 120.423349
3           signal_energy  10.398604
4               duration  11.469464
5          dominant_freq 108.338593
6            brittleness 153.731044
7                 stress  14.807511
8         elastic_energy  14.088887
9   compressive_strength  25.239449
10      tensile_strength  62.532979
```

```
Original features: 11
PCA components after reduction: 10
```

4. **Class Imbalance Handling**

Applied SMOTE (Synthetic Minority Over-sampling Technique) to balance rockburst classes and ensure equal learning across all categories.

```
smote = SMOTE(random_state=42)
X_train_res, y_train_res = smote.fit_resample(X_train, y_train)

✓ 0.0s
```

5. Model Development

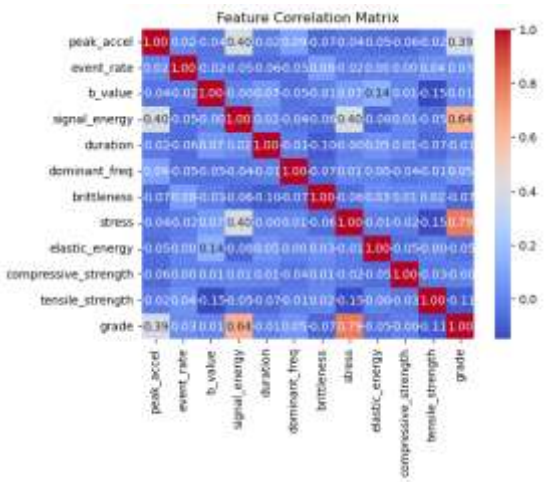
We trained two classical ML models:

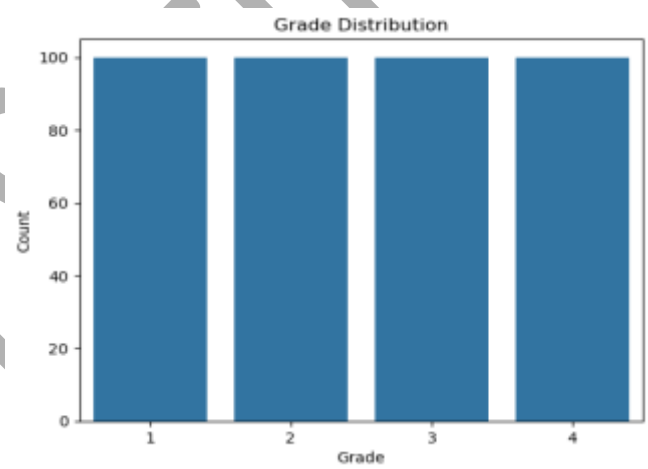
Random Forest – for ensemble learning and feature ranking.

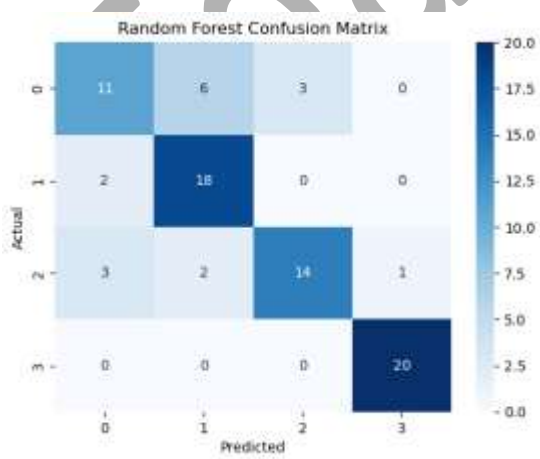
K-Nearest Neighbors (KNN) – for distance-based classification.

Both were trained and validated using:

- PCA-reduced features,
- Train-test split,
- Cross-validation (5-fold),
- Accuracy, F1-score, and confusion matrix.








```
Random Forest Accuracy: 0.7875

Classification Report:
              precision    recall  f1-score   support

           1       0.69      0.55      0.61        20
           2       0.69      0.90      0.78        20
           3       0.82      0.70      0.76        20
           4       0.95      1.00      0.98        20

    accuracy                           0.79        80
   macro avg       0.79      0.79      0.78        80
weighted avg       0.79      0.79      0.78        80
```

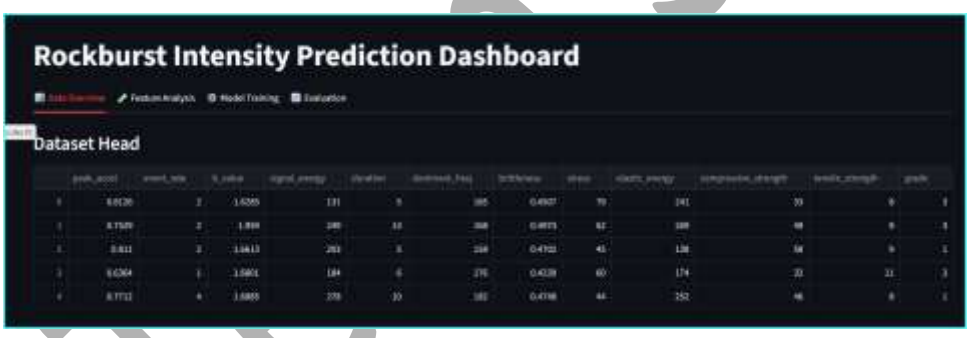
6. Model Saving

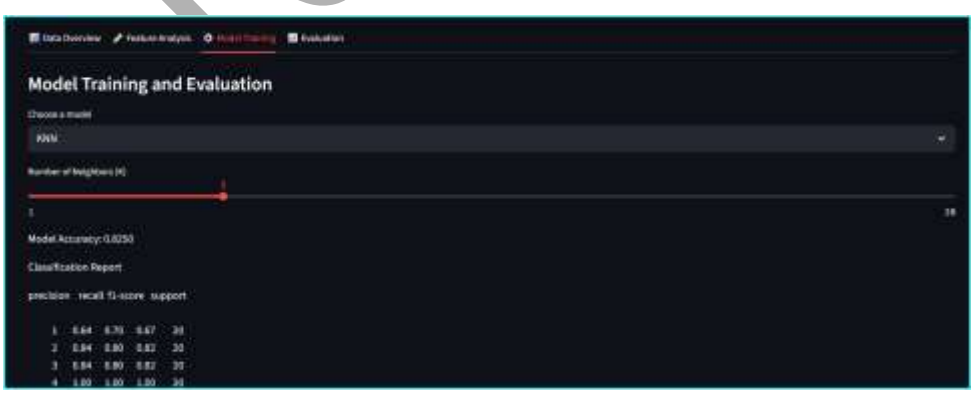
Models were saved using pickle in .pkl format for deployment use.

```
with open("random_forest_model.pkl", "wb") as f:
    pickle.dump(rf_model, f)
with open("knn_model.pkl", "wb") as f:
    pickle.dump(knn, f)
print("Models saved as .pkl files")
```

7. Streamlit Dashboard :

- We developed an interactive Streamlit dashboard for:
- Uploading new data (CSV),
- Making predictions using trained models,
- Displaying model evaluation metrics (accuracy, F1, etc.),
- Showing feature importance from Random Forest,
- Visualizing PCA-reduced data.
- The dashboard makes the entire workflow accessible, interpretable, and user-friendly.





8. Conclusion :

Our pipeline successfully simulates an end-to-end rockburst prediction system using synthetic multimodal data. With proper preprocessing (scaling, PCA), class balancing (SMOTE), model comparison (RF & KNN), and deployment (Streamlit), we demonstrated a practical machine learning workflow for safety-critical mining applications—ready for adaptation to real-world data in the future.

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